

**DEPARTMENT OF
UNIVERSITY INSTITUTE OF COMPUTING
CHANDIGARH UNIVERSITY**



SUBJECT NAME: R PROGRAMMING

PERSONAL FINANCE ANALYZER

SUBJECT CODE: 25CAP-161

SUBMITTED BY:

Mrithul Padinhattayil

25BCD10298

25BCD-1A

D.O. Submission : _____

SUBMITTED TO:

**Mrs. Sushma
(Assistant Professor)**

INDEX

SI No.	Content	Page No.
1	Title Page	0
2	Abstract	1
3	Introduction	1
4	Objectives	2
5	Tools And Technology	2
6	Code With Explanation	3-5
7	Output	5-8
8	Data Description	8-9
9	Conclusion	9
10	Learning Outcomes	10
11	GitHub Link	10



Personal Finance Analyzer

By: Mrithul P

UID: 25BCD10298

ABSTRACT

Managing personal finances effectively is an essential skill, yet many individuals find it difficult to track and analyze their daily expenses. This project presents a **Personal Expense Analysis System** developed using the R programming language, aimed at simplifying the process of expense tracking and providing meaningful insights from financial data. The system allows users to import expense data from a CSV file, clean and process the data, and generate summaries based on categories, dates, and monthly spending.

The project makes use of libraries such as dplyr for data manipulation, ggplot2 for data visualization, and scales for formatting outputs. Key features include category-wise expense analysis, monthly budget comparison, daily spending trends, and cumulative expense tracking. The system also generates visual representations through graphs and exports summarized data for further use.

Overall, this project demonstrates how basic data analysis techniques can be applied to solve real-world problems. It provides a simple, efficient, and beginner-friendly approach to understanding spending habits and encourages better financial decision-making.

INTRODUCTION

Managing personal expenses effectively is an important part of financial discipline, yet it is often overlooked due to lack of proper tools or structured methods. With increasing daily transactions, it becomes difficult to track spending manually or identify clear patterns in expenses.

The **Personal Expense Analysis System** is designed to address this by providing a simple, data-driven approach using R. It processes expense data from a CSV file and organizes it into meaningful summaries based on time and categories. This helps users gain a clearer understanding of their spending behavior.

The project focuses on making analysis easy and accessible through basic data processing and visualizations, enabling users to make more informed financial decisions without requiring advanced technical knowledge.

OBJECTIVES

- To develop a simple system for analyzing personal expense data using R
- To organize and summarize expenses based on categories and time (daily and monthly)
- To compare actual spending with a predefined monthly budget
- To visualize spending patterns using clear and informative graphs
- To identify trends and major areas of expenditure
- To export analyzed data for further use or reporting
- To provide an easy-to-use and beginner-friendly solution for expense management

TOOLS AND TECHNOLOGIES USED

In this project, a few simple tools are used to handle and understand the expense data:

- **R Programming Language**
R is the main language used to build this project. It is useful for working with data, performing calculations, and creating analysis-based programs.
- **dplyr**
This library helps in working with the data easily. It is used to group expenses, calculate totals, and organize the data in a clear way.
- **ggplot2**
This is used to create graphs. It helps turn the data into visual charts like bar graphs and line graphs, which makes it easier to understand spending patterns.
- **scales**
This is a small supporting library that improves how numbers are shown in graphs, such as adding commas to large values for better readability.

Together, these tools help take raw expense data, process it step by step, and present the results in a simple and understandable format.

CODE WITH EXPLANATION

The project is implemented in R using a modular approach, where each function performs a specific task. This improves readability and makes the code easy to maintain.

1. Libraries

```
library(dplyr)
library(ggplot2)
library(scales)
```

These libraries are used for:

- dplyr: data manipulation (grouping, summarizing)
- ggplot2: creating graphs
- scales: formatting numbers in graphs

2. Output Directory Setup

```
get_script_path <- function() { ... }
output_dir <- get_script_path()
```

This part determines the folder where the script is located and saves all output files in the same directory.

3. Theme Customization

```
theme_modern <- function() { ... }
```

A custom theme is created to give all graphs a clean and modern appearance.

4. Data Loading and Cleaning

```
load_data <- function() { ... }
```

- Prompts the user to select a CSV file
- Converts Date into proper format
- Converts Amount into numeric
- Checks for invalid entries

5. Monthly Summary

```
monthly_summary <- function(data, budget) { ... }
```

- Groups data by month
- Calculates total spending
- Compares with budget

6. Category Analysis

```
category_analysis <- function(data) { ... }
```

- Groups by category
- Calculates total spending
- Sorts results

7. Visualization Functions

```
plot_category <- function(data) { ... }
```

```
plot_monthly <- function(data, budget) { ... }
```

```
plot_daily <- function(data) { ... }
```

```
plot_cumulative <- function(data) { ... }
```

These functions generate different charts for analysis.

8. Export Functions

```
export_category <- function(data) { ... }
```

```
export_monthly <- function(data, budget) { ... }
```

These save results as CSV files.

9. Main Function

```
run_analysis <- function(budget = 30000) { ... }
```

Controls the full workflow of the program.

10. Execution

```
run_analysis("SET YOUR BUDGET")
```

```
run_analysis(30000)
```

Runs the program with a default budget.

RESULTS AND OUTPUT

The system successfully processes the input expense data and generates both numerical summaries and visual outputs. These results help in understanding spending patterns in a clear and structured manner.

Key Outputs

- **Monthly Summary:**
Displays total spending for each month along with the assigned budget. It also shows whether the user is within budget or has exceeded it.
- **Category-wise Analysis:**
Provides a breakdown of total expenses for each category, helping identify where the most money is being spent.
- **Top Expenses:**
Lists the highest individual transactions, giving insight into major expenditures.


```

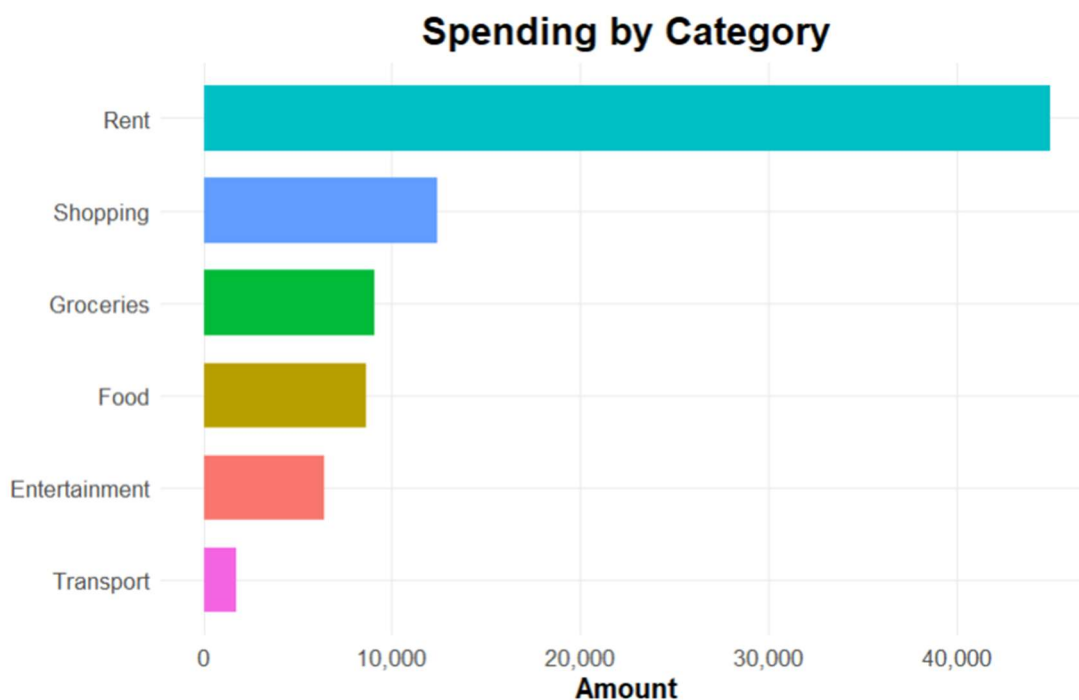
Summary
Total Spending: 83200
Transactions: 39

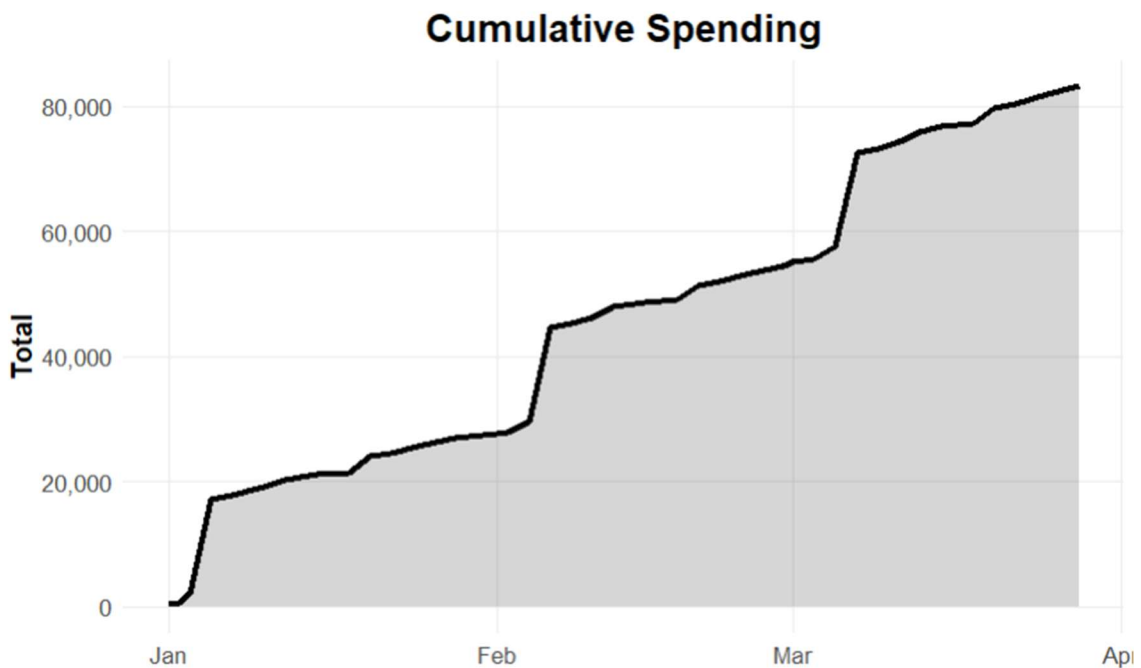
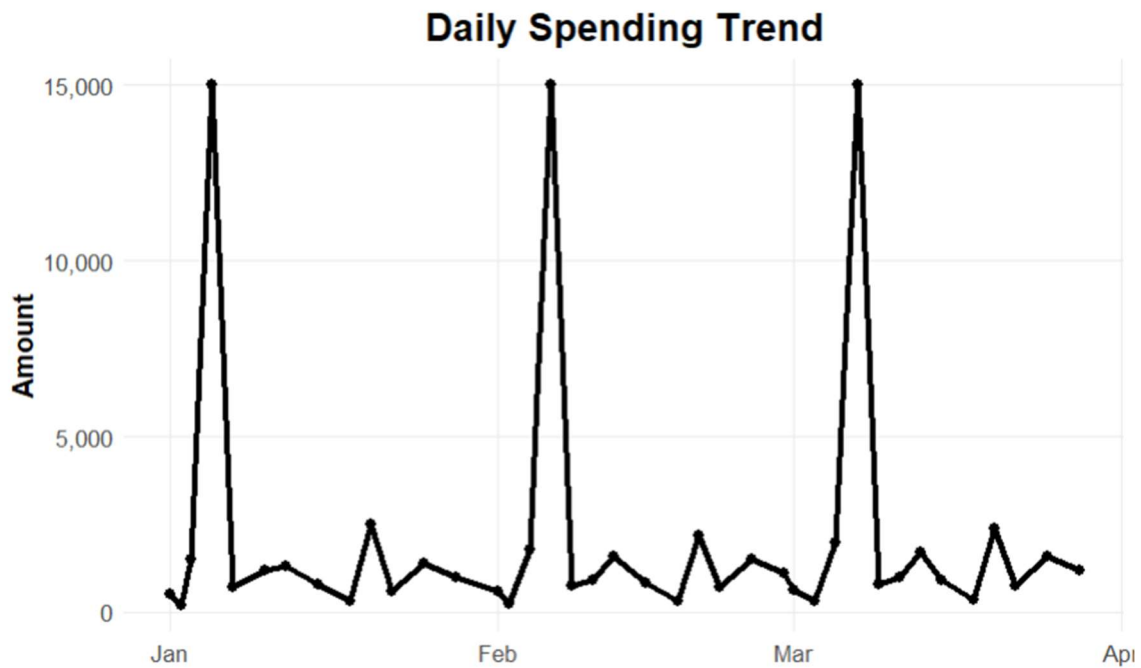
Monthly
# A tibble: 3 × 5
  Month   Total Budget Remaining Status
  <chr>   <dbl> <dbl>   <dbl> <chr>
1 2025-01 27000  30000     3000 Within Budget
2 2025-02 27550  30000     2450 Within Budget
3 2025-03 28650  30000     1350 Within Budget

Top Expenses
  Date Category Amount
1 2025-01-05 Rent 15000
2 2025-02-06 Rent 15000
3 2025-03-07 Rent 15000
4 2025-01-20 Shopping 2500
5 2025-03-20 Shopping 2400
# A tibble: 3 × 5
  Month   Total Budget Remaining Status
  <chr>   <dbl> <dbl>   <dbl> <chr>
1 2025-01 27000  30000     3000 Within Budget
2 2025-02 27550  30000     2450 Within Budget
3 2025-03 28650  30000     1350 Within Budget

```

Visual Outputs





Insights

From the generated outputs, users can:

- Identify their highest spending categories
- Detect months where they exceeded their budget
- Understand overall spending trends

- Make better financial decisions based on data

Overall, the system provides meaningful insights by combining data analysis with visual representation, making it easier to interpret and use the results effectively.

DATA DESCRIPTION

The dataset used in this project is stored in a CSV file with three main columns: **Date**, **Category**, and **Amount**. Each row represents a single expense entry recorded on a specific date.

- **Date** – Indicates when the expense occurred (format: YYYY-MM-DD)
- **Category** – Describes the type of expense, such as Food, Transport, Shopping, Rent, Entertainment, or Groceries
- **Amount** – Represents the money spent for that particular entry (in currency units)

The dataset contains multiple entries across different dates, allowing analysis of spending patterns over time. Some categories, like **Food** and **Transport**, appear multiple times, making it possible to calculate totals and compare expenses across categories.

Overall, the data is structured in a simple tabular format, making it easy to process, analyze, and visualize using programming tools.

Sample Dataset:

Date,Category,Amount

2025-01-01,Food,500

2025-01-02,Transport,200

2025-01-03,Shopping,1500

2025-01-05,Rent,15000

2025-01-07,Food,700

2025-01-10,Entertainment,1200

2025-01-12,Groceries,1300

2025-01-15,Food,800

2025-01-18,Transport,300

2025-01-20,Shopping,2500

2025-01-22,Food,600

2025-01-25,Groceries,1400
2025-01-28,Entertainment,1000

2025-02-01,Food,600
2025-02-02,Transport,250
2025-02-04,Shopping,1800
2025-02-06,Rent,15000
2025-02-08,Food,750
2025-02-10,Entertainment,900
2025-02-12,Groceries,1600
2025-02-15,Food,850
2025-02-18,Transport,300
2025-02-20,Shopping,2200
2025-02-22,Food,700
2025-02-25,Groceries,1500
2025-02-28,Entertainment,1100

CONCLUSION

The Personal Expense Analysis System provides a simple and effective way for users to track, analyze, and visualize their spending habits. By using a structured CSV input, the system quickly processes data to generate monthly and category-wise summaries, highlights top expenses, and produces clear visualizations of daily, monthly, and cumulative spending.

This project demonstrates how basic programming and data analysis techniques in R can be applied to solve a real-world problem. It empowers users to better understand their financial behavior, manage their budget, and make informed decisions about their spending.

LEARNING OUTCOMES

- Gained practical experience in handling and analyzing real-world data using programming.
- Learned how to import, clean, and process data from CSV files effectively.
- Developed an understanding of performing basic data analysis such as calculating totals and averages.
- Improved skills in data visualization by creating graphs and charts for better interpretation.
- Enhanced knowledge of using tools and libraries to build a complete, functional data analysis project.
- Strengthened problem-solving and logical thinking through implementing the project step by step.

GITHUB LINK

🔗 <https://github.com/MrithulP/Personal-Finance-Analyzer>